Total Printed Pages: 04

Roll No:

CSA-101

Examination –Dec- 2023

B.Tech. I Sem : CS, IT, AI, BC, IoT,

Introduction to Computer Science and Engineering

Time : 2 Hrs

Max. Marks : 60
Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.
Word limit be observed as follows:
Part a & b – Objective type/Fill in the blanks/One sentence answer.
Part c – Max 100 words and Part d – Max 400 words.
Word limit NOT to be followed for diagram, numerical, derivation.

			Marks	BL	CO
Q.1	(a)	(i)..... is the most powerful computer made and physically they are some of the largest. (ii)..... has the largest capacity of all the optical media available and go up to 100 GB at present.	02	1	1
	(b)	(i) .....has read/write heads that allows data to be written to or read from the disk surface. (ii) ..... is a very thin strip of plastic coated in a magnetic layer read and written by a read/write head.	02	1	1
	(c)	Explain the Computer organization with diagram.	03	1	2
		OR			
		What is Instruction Set?	03	1	2
	(d)	Describe in brief the Generation of Computers.	05	2	2
		OR			
		How does Memory hierarchy increase the performance of Computer storage? Explain It.	05	2	2

Q.2	(a)	#include <stdio.h>
		int main() { float q = 'a';
		printf("%f", q);
		return 0;
		}
		(a) run time error (b) a (c) 97.000000 (d) a.0000000
			02	1	2

	(b)	What will be the output of the following C code?
		#include<stdio.h>
		int main()
		{
		int p = 1, q = 2, r = 3, s = 4, x;
		e = r + s = q * p;
		printf("%d, %d\n", x, s);
		}
		(a) Syntax error (b) 5, 2 (c) 7, 2 (d) 7, 4
			02	1	1

	(c)	Explain a general structure of C program with an example.	03	2	2
		OR			
		What is a token? What are different types of tokens available in C language? Explain.	03		
	(d)	Show how break and continue statements are used in a C-program, with example.	05	3	1
		OR			
		Develop a C program to generate and plot the Pascal triangle.	05		
Q.3	(a)	(i) How many times the program will print "Computer"?	02	1	3

```
int main() {
 printf("Computer");
 main();
 return 0;
}
```

(a) Infinite times (b) 32767 times (c) 65535 times (d) Till stack overflows

(ii) What is the output of C program with pointers?

```
int main() {
 int a = 20;
 //a memory location = 1234
 printf("%d %d %d %d", a, &a, *(&a));
 return 0;
}
```

(a) 20 20 20 (b) 20 1234 1234 (c) 20 1234 20 (d) 20 20 20

	(b)	(i) What is the output of C program with recursive function?	02	1	3

```
int sum(int x) {
 int k = 1
 if(x <= 1)
  return 1;
 k = x + sum(x - 1);
  return k;
}
```

(a) 10 (b) 11 (c) 12 (d) 15

(ii) A recursive function can be replaced with __ in C language.

(a) for loop (b) while loop (c) do while loop (d) All the above

	(c)	Show how break and continue statements are used in a C program, with example	03	2	1
		OR			
		What is array? Explain the declaration and initialization of one dimensional and two dimensional array with an example	03	2	1

→ Diff b/w call by value and call by reference

		frequency of each of the vowels and total count of consonants?			
		OR			
		What is Recursion? Write a C program to compute Factorial of a given number using recursion.	05	2	1

Q.4	(a)	(i) What is the base data type of a pointer variable by which the memory would be allocated to it?
		(a) int (b) float (c) No datatype (d) unsigned int
		(ii) What is the output of the following C code?
		`char *ptr;`
		`char mystring[] = "abcdefg";`
		`ptr = myString;`
		`ptr += 5;`
		(a) fg (b) efg (c) defg (d) cdefg
			02	02	1

	(b)	(i) What will be the size of the following structure?
		`#include <stdio.h>`
		`struct temp {`
		`int a[10];`
		`char p;`
		`};`
		(a) 5 (b) 11 (c) 20 (d) 21
		(ii) Choose a correct statement about C structure elements?
		(a) Structure elements are stored on random free memory locations
		(b) structure elements are stored in register memory locations
		(c) structure elements are stored in contiguous memory locations
		(d) None of the above
			02	02	4

	(c)	What is structure? Explain the C syntax of structure declaration with example	03	03	4
		OR			
		Explain the difference between array and structures.	03	03	4
	(d)	Explain the array of pointes with example? or Explain how pointers and arrays are related with example.	05	5	2
		OR			
		What is preprocessor directive? Explain #define and #include preprocessor directives	05	5	2

Q.5	(a)	(i) What is a computer network? (a) A device used to display information on a computer screen (b) A collection of interconnected computers and devices that can communicate and share resources (c) A type of software used to create documents and presentations (d) The physical casing that protects a computer's internal components (ii) Which of the following is an example of Bluetooth? (a) wide area network (b) virtual private network (c) local area network (d) personal area network	02	1	5
	(b)	(i) What is Machine learning? (a) The autonomous acquisition of knowledge through the use of computer programs (b) The autonomous acquisition of knowledge through the use of manual programs (c) The selective acquisition of knowledge through the use of computer programs (d) The selective acquisition of knowledge through the use of manual programs (ii) What is HTML? (a) HTML describes the structure of a webpage (b) HTML is the standard markup language mainly used to create web pages (c) HTML consists of a set of elements that helps the browser how to view the content (d) All of the mentioned	02	3	4
	(c)	What is Operating System? Write various types.	03	2	5
		OR			
		Explain Data Science.	03	2	5
	(d)	How cloud computing services ease the service based facilities?	05	2	5
		OR			
		Why does Crypto currency use the block chain Technology?	05	2	5

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

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